

Microstrip Ring Resonator on Polyimide: Model Development and Modification Guide

openEMS Ring Resonator Dk/Loss Extraction Model

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Contents

1 Purpose	1
2 Model Development History	1
3 Model Structure	3
3.1 Geometry overview	3
3.2 Key parameters in the script	3
3.3 Trace width synthesis	3
3.4 Ring sizing and the resonance condition	4
3.5 Dk and loss-tangent extraction	4
4 Limitations and Interpretation Notes	4
5 Running the Model	5
6 Modifying the Model	5
6.1 Changing the dielectric properties	5
6.2 Changing the ring/feed geometry	5
6.3 Adding more substrate layers	6
7 File Summary	7

1 Purpose

This document describes the openEMS/CSXCAD full-wave FDTD model implemented in `07_ring_resonator_poly`. The model simulates a microstrip ring resonator, weakly gap-coupled to a single straight through feed line, fabricated on a thin polyimide substrate. A single wideband Gaussian excitation spanning 100 MHz–67 GHz is used so that every harmonic resonance of the ring appears as a notch in the transmitted S-parameter $|S_{21}|$. From the measured resonant frequencies, -3 dB bandwidths, and insertion losses, the script extracts the substrate’s relative permittivity (Dk) and loss tangent ($\tan \delta$) as a function of frequency — the classic *ring-resonator method* used for microwave/mm-wave laminate characterization.

This is a practical, general-purpose starting point: the same script structure can be adapted to other substrates, stack-ups, and ring/feed geometries by editing a small number of parameters near the top of the file, as described in Section 6.

2 Model Development History

The model went through several rounds of debugging before it converged on physically correct S-parameters. These are documented here because the underlying causes are generic FDTD/microstrip modeling pitfalls that are easy to reintroduce when the geometry or substrate stack is modified – Section 6 calls out where each rule must be respected again.

1. **Absorbing boundary choice.** The MUR boundary condition was found to be numerically unstable (unbounded exponential energy growth) for this broadband (100 MHz–67 GHz) case, especially in combination with a PEC ground plane sitting exactly on the `zmin` domain boundary. The fix was to use PML_8 on all open boundaries and PEC only on the `zmin` face (where the ground plane already terminates the field):

```
1 FDTD.SetBoundaryCond(["PML_8","PML_8","PML_8","PML_8","PEC","PML_8"])
```

2. **Port margin.** A lumped port's current probe measures a closed H-field loop that needs a valid FDTD cell on *both* sides of the probe. A port placed exactly on the outermost mesh line of the simulation domain has no cell on its outward side, so the measured current is pure numerical noise, corrupting S_{11}/S_{21} even though the simulation itself remains stable. The substrate and ground plane are extended by a `margin` (1 mm) beyond each port so the ports sit a few cells inside the domain.
3. **Feed line must stop exactly at the port planes.** The metal trace must *not* extend into the margin region past $x = 0$ or $x = \text{sub}_1$: a continuous metal sheet running straight over a lumped port's vertical probe shorts out the port's series resistor, again causing near-total, frequency-independent reflection.
4. **Open-boundary air gap (z-direction).** Without sufficient air space above the trace, the PML/`zmax` boundary sits right on (or a fraction of a millimeter above) the metal, which either silently falls back to PEC behavior (too few z -lines for a full 8-cell PML) or absorbs the near field before it can couple properly. The fix grades the z -mesh manually above the trace with a geometrically growing sequence (ratio 1.3) out to `z.air = max(10h, 5w, 1.5)` mm of clearance.
5. **Lateral clearance (y-direction).** The same principle applies in-plane: the feed line's outer edge must be kept well clear of the PML_8 boundary at $y = 0$. A microstrip's fringing E-fields extend several substrate-heights beyond the trace edge; truncating that too close to an absorbing boundary parasitically loads the line (acts like a nearby ground wall), severely distorting its characteristic impedance and causing large, frequency-independent reflection at *every* frequency – including very low ones where a matched line should behave almost ideally. The fix introduces

```
1 y_clear = max(10 * h, 5 * w, 0.5) # [mm] lateral clearance to y=0 boundary
```

and uses it to size the substrate width and feed-line offset.

6. **Mesh-line deduplication vs. exact port boundaries.** A deduplication pass (`dedupe_lines`) merges near-coincident mesh lines to avoid tiny sliver cells that would otherwise force an unstable CFL time step. This can silently merge an exact, manually inserted port-boundary line into a nearby fine-grid point, drifting the port's edge away from the intended trace edge. The fix re-inserts the exact port-boundary lines (`mesh.AddLine("x", [0, sub_1])`) and the y edges of the feed line) *after* the dedup pass.

7. **Kappa reference frequency.** openEMS’s `kappa` material parameter is a plain (frequency-independent) Ohmic conductivity, so the *effective* loss tangent it produces scales as $\tan \delta_{\text{eff}}(f) = \tan \delta_{\text{nominal}} \cdot f_0/f$ – it is **not** constant with frequency. Using the excitation’s arithmetic-mean center frequency as f_0 badly overestimates loss at the low end of a log-wide band. The fix uses the *geometric* mean of the band edges instead:

```

1 f0_kappa = float(np.sqrt(f_start * f_stop))
2 kappa = tanD_nominal * 2 * np.pi * f0_kappa * er_nominal * eps0

```

This minimizes the worst-case log-frequency error across the band but does not eliminate it – see Section 4.

8. **Resonance signature: dips, not peaks.** For this single-tap, gap-coupled ring topology, each ring resonance couples energy *out of* the through line and into the ring, producing a notch (local minimum) in $|S_{21}|$ at resonance, not a peak. The post-processing peak finder searches for local minima, not maxima.
9. **Harmonic-index assignment.** The lowest-order ring harmonics are the most weakly coupled and can be too shallow to cross the detection threshold, especially before the simulation has fully converged. If the detected notches were simply numbered $n = 1, 2, 3, \dots$ in ascending frequency order, a missing low-order harmonic would silently mislabel every harmonic above it. The script instead estimates n directly from the physical resonance condition (Eq. 1) using the nominal effective permittivity as an initial guess, then rounds to the nearest integer.

3 Model Structure

3.1 Geometry overview

Element	Description
Ground plane	Full-size PEC sheet at $z = 0$, covering the substrate footprint (plus margin)
Substrate	Single dielectric slab, thickness h , from $z = 0$ to $z = h$
Feed line	Straight microstrip trace of width w , thickness t , at $z \in [h, h + t]$
Ring	Cylindrical-shell (annulus) trace, mean radius r_{mean} , width w , same z range
Ports	Two lumped $50 \, \Omega$ ports, one at each end of the feed line (z -directed)

The ring is placed alongside the feed line, separated by a coupling gap `gap`, so that the ring’s outer edge is a distance `gap` from the feed line’s edge (edge-coupled, not tap-coupled). Only a single feed line couples to the ring (a “one-port-pair, single-tap” topology): energy is launched at port 1, mostly transmitted straight through to port 2, with a small fraction resonantly coupled into the ring at each of its harmonic frequencies.

3.2 Key parameters in the script

Variable	Meaning	Default
<code>f_start, f_stop</code>	Simulation frequency band	0.1–67 GHz
<code>er_nominal</code>	Substrate relative permittivity (Dk)	3.5
<code>tanD_nominal</code>	Substrate loss tangent	0.008
<code>h</code>	Substrate thickness [mm]	0.1
<code>w</code>	Trace width [mm], synthesized for 50 Ω	(computed)
<code>trace_t</code>	Copper trace thickness [mm]	0.017 (0.5 oz)
<code>f1_target</code>	Target fundamental ring resonance	8.5 GHz
<code>r_mean</code>	Ring mean radius [mm]	(computed from <code>f1_target</code>)
<code>gap</code>	Ring-to-feed-line coupling gap [mm]	0.12
<code>feed_len</code>	Feed stub length each side [mm]	4.0
<code>margin</code>	Port/substrate margin past the ports [mm]	1.0
<code>y_clear</code>	Lateral clearance, trace to PML [mm]	$\max(10h, 5w, 0.5)$
<code>z_air</code>	Vertical clearance, trace to PML [mm]	$\max(10h, 5w, 1.5)$

3.3 Trace width synthesis

The feed-line/ring trace width w is synthesized for a 50 Ω characteristic impedance using the closed-form Hammerstad synthesis formula, given the substrate’s nominal ε_r and thickness h :

$$A = \frac{Z_0}{60} \sqrt{\frac{\varepsilon_r + 1}{2}} + \frac{\varepsilon_r - 1}{\varepsilon_r + 1} \left(0.23 + \frac{0.11}{\varepsilon_r} \right), \quad \frac{W}{h} = \frac{8e^A}{e^{2A} - 2} \quad (W/h < 2).$$

This is only valid for the narrow-trace ($W/h < 2$) regime; for thicker substrates or higher-Dk materials where the synthesized $W/h \geq 2$, the wide-trace Hammerstad formula should be substituted instead (see Section 6).

3.4 Ring sizing and the resonance condition

The ring’s mean radius is chosen so its fundamental ($n = 1$) resonance sits near a target frequency `f1_target` (default 8.5 GHz), spreading roughly seven harmonics across the 100 MHz–67 GHz simulation band. The governing relation is the ring’s circumferential standing-wave condition:

$$n \lambda_g(f_n) = 2\pi r_{\text{mean}} \implies \varepsilon_{\text{eff},n} = \left(\frac{n c_0}{2\pi r_{\text{mean}} f_n} \right)^2, \quad (1)$$

where c_0 is the speed of light in vacuum and f_n is the n -th measured resonant frequency. This same relation, inverted, is used both to size the ring initially (given a nominal ε_{eff}) and to recover $\varepsilon_{\text{eff},n}$ from each measured resonance.

3.5 Dk and loss-tangent extraction

For each detected $|S_{21}|$ notch:

1. Determine f_n and its harmonic index n (Eq. 1).
2. Compute $\varepsilon_{\text{eff},n}$ from Eq. 1.

3. Invert the Hammerstad–Jensen effective-permittivity relation for the known (fixed) trace W/h to recover $\varepsilon_{r,n}$ (Dk):

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} F(W/h), \quad F(W/h) = \left(1 + \frac{12}{W/h}\right)^{-1/2}.$$

4. Measure the loaded $Q_L = f_n/\text{BW}_{3\text{dB}}$ from the notch’s -3 dB bandwidth, then the unloaded Q_u via the standard insertion-loss correction $Q_u = Q_L/(1 - 10^{-\text{IL}_{\text{dB}}/20})$.
5. Compute an analytic conductor-loss-only Q_c from the Wheeler incremental-inductance rule and copper surface resistance, and remove it from Q_u to isolate the dielectric-only Q_d : $1/Q_d = 1/Q_u - 1/Q_c$ (only when $Q_c > Q_u$; otherwise Q_u itself is reported as an upper bound on Q_d , see Section 4).
6. Recover the loss tangent via the dielectric filling factor $q = (\varepsilon_{\text{eff}} - 1)/(\varepsilon_r - 1)$:

$$\tan \delta_n = \frac{1}{Q_{d,n} q_n}.$$

4 Limitations and Interpretation Notes

- **Convergence:** The default `NrTS = 200000` / `EndCriteria = 1e-5` (-50 dB) budget may not be reached for a genuinely high- Q ring before the timestep budget is exhausted (openEMS will print a warning and stop at `NrTS`). An incompletely decayed time-domain signal narrows the apparent resonance linewidth (spectral leakage), which *inflates* the apparent Q_u and therefore *underestimates* the extracted $\tan \delta$. If the extracted Dk looks reasonable but $\tan \delta$ looks too low (or increases smoothly as frequency decreases across the band), increase `NrTS` and/or relax `EndCriteria` slightly and re-run.
- **Frequency-independent kappa:** because openEMS models a constant Ohmic conductivity rather than a true frequency-independent $\tan \delta$, the simulated substrate loss is only exactly correct at $f = f_0_kappa$ (the geometric mean of the band) and drifts (over- or under-representing the true material loss) away from it. This is a modeling compromise, not a bug; it does not need to be “fixed” for Dk extraction (which does not depend on kappa), but should be kept in mind when interpreting the extracted $\tan \delta$ trend vs. frequency.
- **Analytic conductor-loss model:** the built-in Q_c estimate is a simplified microstrip attenuation formula, not a full analytic model (it ignores edge current crowding, surface roughness, etc.). If Q_c comes out lower than the measured Q_u (physically impossible for a clean loss-mechanism separation), the script falls back to reporting Q_u directly, which is a conservative (upper-bound-on-Dk-loss, lower-bound-on-actual- $\tan \delta$) estimate rather than a hard failure.

5 Running the Model

```
1 conda activate openems
2 python 07_ring_resonator_polyimide_dk_loss.py
```

Simulation results (port time/frequency-domain data, and the final S-parameter/Dk/loss plot `ring_resonator_dk_loss.png`) are written to `sim_results/ring_resonator/`. A full run at the default mesh density takes on the order of a few hours of wall-clock time on a modern multi-core workstation (roughly 2.8 million FDTD cells, tens of thousands of timesteps).

6 Modifying the Model

All of the changes below only require editing parameters/geometry near the top of `07_ring_resonator_polyimide.c`. The mesh, port, and boundary-condition logic further down the file automatically adapts (re-derives clearances, resolutions, etc.) from these parameters and does not need to be touched for routine changes described here.

6.1 Changing the dielectric properties

To model a different substrate material, edit only:

```
1 er_nominal = 3.5 # relative permittivity (Dk)
2 tanD_nominal = 0.008 # loss tangent
3 h = 0.1 # substrate thickness [mm]
```

Everything else (trace width synthesis, ring sizing, mesh resolution, kappa) is automatically re-computed from these three values. If the new material's synthesized trace width comes out with $W/h \geq 2$ (thick, low-Dk substrates, or very thin high-Dk ones), replace the narrow-trace Hammerstad formula (Section 6, trace-width synthesis) with the wide-trace variant:

$$\frac{W}{h} = \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left(\ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right], \quad B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}.$$

6.2 Changing the ring/feed geometry

- **Fundamental resonance frequency:** change `f1_target` (default 8.5 GHz). The ring radius `r_mean` is automatically recomputed from Eq. 1 using the nominal ϵ_{eff} , so the harmonic comb shifts accordingly. Keep in mind the number of harmonics visible in the fixed 100 MHz–67 GHz band scales roughly as $67 \text{ GHz}/f_1$.
- **Coupling strength:** change `gap` (default 0.12 mm). Smaller gaps couple more strongly (deeper, wider notches, lower loaded Q_L); larger gaps couple more weakly (shallower notches that may fall below the detection threshold, particularly at low frequency – see Section 4).
- **Feed line length:** change `feed_len` (default 4.0 mm each side). This mainly affects total simulation domain size/run time and has negligible effect on the extracted results as long as it stays large enough that the port's near-field has settled before reaching the ring.
- **Trace width / impedance:** the trace width `w` is automatically synthesized for 50Ω from `er_nominal` and `h`. To target a different characteristic impedance, change `Z0` (default 50.0) before the synthesis block.
- **Copper thickness:** change `trace_t` (default 0.017 mm, i.e. 0.5 oz copper) to model a different copper weight.

Whenever the ring radius, gap, or trace width changes substantially, sanity check that the automatically-derived clearances (`y_clear`, `z_air`, `margin`) are still being applied correctly (Section 6 and history items 4 and 5): these are all defined as functions of `h` and `w`, so they should scale appropriately on their own, but always re-inspect the printed geometry summary at the start of a run to confirm reasonable values before committing to a multi-hour simulation.

6.3 Adding more substrate layers

The current model uses a single dielectric slab between the ground plane and the trace. To model a multilayer stack-up (e.g., a core + prepreg, or a polyimide film laminated to a carrier), replace the single `CSX.AddMaterial("Polyimide", ...)` block with one material and one `AddBox` call per layer, stacked in z :

```
1 # Example: two-layer substrate (bottom layer + top layer under the trace)
2 h1, er1, tanD1 = 0.05, 3.5, 0.008 # bottom layer (polyimide)
3 h2, er2, tanD2 = 0.05, 4.2, 0.015 # top layer (e.g. bond-ply/adhesive)
4 h = h1 + h2 # total substrate thickness used everywhere else
5
6 kappa1 = tanD1 * 2 * np.pi * f0_kappa * er1 * eps0
7 kappa2 = tanD2 * 2 * np.pi * f0_kappa * er2 * eps0
8
9 sub1 = CSX.AddMaterial("Polyimide_bottom", epsilon=er1, kappa=kappa1)
10 sub1.AddBox(priority=0, start=[-margin, 0, 0], stop=[sub_l + margin, sub_w, h1])
11
12 sub2 = CSX.AddMaterial("Polyimide_top", epsilon=er2, kappa=kappa2)
13 sub2.AddBox(priority=0, start=[-margin, 0, h1], stop=[sub_l + margin, sub_w, h1 + h2])
14
15 # Ground plane, feed line, ring, and ports are unaffected: they already
16 # reference z=0 (ground) and z=h..h+trace_t (trace) generically, so as
17 # long as 'h' above is redefined as the TOTAL stack thickness, no other
18 # geometry code needs to change.
```

Important considerations when adding layers:

- **Effective permittivity synthesis** (trace width, ring sizing, $F(W/h)$ inversion) assumes a single homogeneous substrate. For a true multilayer stack, either (a) use a weighted/parallel-plate approximate ϵ_r for the synthesis step only (reasonable when one layer dominates the field energy, e.g. a thin bond-ply under a much thicker core), or (b) treat `er_nominal` in the synthesis formulas as an initial guess and rely on the FDTD-measured $\epsilon_{\text{eff},n}$ per layer stack as the authoritative result – the resonance-based extraction in Section 3.4 works correctly regardless of how many dielectric layers are present, since it only depends on the *measured* resonant frequencies, not on the analytic synthesis formulas used to size the geometry initially.
- **Mesh:** add a z -mesh line at every layer interface (`mesh.AddLine("z", [0, h1, h1+h2, ...])`) so each layer boundary is resolved exactly; the existing fine-to-coarse grading logic for the open-boundary air gap above the trace does not need to change, since it already starts counting from `h + trace_t`, the top of the (now multi-layer) stack.
- **Reporting:** if you want per-layer Dk/loss reported separately rather than a single lumped “Dk” number, you will need a more detailed post-processing model (e.g. conformal-mapping partial-capacitance methods) since the ring-resonator method as implemented here only recovers a single effective ϵ_r per resonance, consistent with a single equivalent homogeneous substrate.

7 File Summary

File	Description
07_ring_resonator_polyimide_dk_loss.py	Full model: geometry, mesh, simulation, post-processing
sim_results/ring_resonator/	openEMS output directory (port data, plot)
ring_resonator_dk_loss.png	Final S-parameter / extracted Dk / extracted $\tan \delta$ plot